

Requirements Engineering – Lab 1

Problem Statement #34: Digital Gift Card & Loyalty Point Exchange

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Requirements Table

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: A valid point conversion produces a single-use voucher with a cryptographic checksum. Fail: An expired or duplicate voucher is redeemed.	Ensures valid loyalty-point conversion and secure gift-card redemption.
FR-002	Functional	The system shall allow an account holder to aggregate loyalty points from supported	High	Pass: Points from a supported merchant are successfully added to the account holder's	Enables users to combine loyalty points from multiple supported merchants.

		merchant partners.		loyalty balance. Fail: Points from an unsupported merchant are added.	
FR-003	Functional	The system shall calculate universal exchange credits using the current dynamic conversion rate for the user's loyalty points.	High	Pass: The system converts the user's loyalty points using the currently applicable conversion rate and displays the resulting exchange credits. Fail: The system uses an outdated or incorrect conversion rate	Ensures loyalty points are converted accurately according to dynamic exchange rates.
FR-004	Functional	The system shall allow an account holder to redeem eligible digital gift cards using universal exchange credits.	High	Pass: A gift card is issued when the account holder has sufficient exchange credits. Fail: Redemption is rejected when the account holder has insufficient exchange credits.	Provides the core gift-card redemption functionality of the platform.
FR-005	Functional	The system shall lock a digital voucher during redemption and prevent its duplicate or expired redemption.	High	Pass: A voucher is locked during redemption and cannot be redeemed again. Fail: An expired or previously redeemed	Prevents fraudulent and duplicate gift-card redemption.

				voucher is accepted.	
NFR-001	Nonfunctional	Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: Benchmarking tests confirm that point conversion and gift card redemption meet the 250 ms target with zero ledger balance drift under simulated peak load.	Ensures fast processing and accurate account balances during high usage.
NFR-002	Nonfunctional	The system shall ensure that each generated digital voucher can be redeemed only once and shall reject expired or previously redeemed vouchers.	High	Pass: A valid voucher is accepted once and an expired or previously redeemed voucher is rejected. Fail: The system accepts a duplicate or expired voucher.	Protects the platform against duplicate and fraudulent voucher redemption.